

8	The Earth takes 24 hours to complete one rotation on its axis. What is the angular velocity of the Earth as it rotates on its axis?		
	<table border="1"> <tr> <td data-bbox="344 319 467 396">A</td><td data-bbox="467 319 2489 396">$1.75 \times 10^{-3} \text{ rad s}^{-1}$</td></tr> </table>	A	$1.75 \times 10^{-3} \text{ rad s}^{-1}$
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	<table border="1"> <tr> <td data-bbox="344 396 467 474">B</td><td data-bbox="467 396 2489 474">$1.99 \times 10^{-7} \text{ rad s}^{-1}$</td></tr> </table>	B	$1.99 \times 10^{-7} \text{ rad s}^{-1}$
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	<table border="1"> <tr> <td data-bbox="344 474 467 552">C</td><td data-bbox="467 474 2489 552">$4.36 \times 10^{-3} \text{ rad s}^{-1}$</td></tr> </table>	C	$4.36 \times 10^{-3} \text{ rad s}^{-1}$
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	Answer: D $\omega = 2\pi / T$ $= 2\pi / 24 \times 60 \times 60$ $= 7.27 \times 10^{-5} \text{ rad s}^{-1}$		

9	A stone is attached to a string. The stone is then caused to swing in a vertical circular motion at
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